

Session – 1

Q1) Name 5 IT job roles and write one line about what each role does.

Ans: Software Developer – Develops, codes, tests, and maintains software applications and websites.

Software Tester (QA Engineer) – Tests software to find bugs and ensures the application works correctly and meets requirements.

UI/UX Designer – Designs the look, feel, and user experience of websites and applications.

Data Analyst – Analyzes data to identify trends and provides insights that help businesses make better decisions.

AI Engineer – Builds and integrates artificial intelligence and machine learning systems to solve real-world problems.

Q2) List 3 product companies and 3 service companies with one example of work each does.

Ans: Product Companies

1. Google – Builds products like Google Search, Gmail, YouTube, and Google Maps for millions of users.

2. Microsoft – Develops products such as Windows, Microsoft 365, and Azure cloud services.

3. Zoho – Develops business software products such as Zoho CRM, Zoho Books, and Zoho Projects.

--Service Companies

1. TCS (Tata Consultancy Services) – Provides IT services such as developing and maintaining software solutions for client companies.

2. Infosys – Provides software development, consulting, cloud, and digital transformation services to clients.

3. Wipro – Provides IT consulting, application development, cloud, and technology services to businesses.

Q3) Write the full forms and one-line use of: GitHub, Slack, Jira, IDE, API.

Ans: 1. GitHub

Full Form: GitHub does not have a specific full form (it is named after "Git," which is a version control system).

Use: It is a platform where developers can store their code online, track changes, and collaborate on projects together.

2. Slack

Full Form: Searchable Log of All Conversation and Knowledge.

Use: It is a messaging app used by professional teams to communicate, share files, and organize their work conversations.

3. Jira

Full Form: Jira does not have a full form (the name is actually derived from "Gojira," the Japanese word for Godzilla).

Use: It is a project management tool used by software teams to plan tasks, assign work, and track bugs or issues.

4. IDE

Full Form: Integrated Development Environment.

Use: It is a software application that provides programmers with all the tools they need to write, edit, and test code in one single place.

5. API

Full Form: Application Programming Interface.

Use: It acts as a bridge or messenger that allows two different software programs to talk to each other and share data.

Q4) Name any 5 programming languages and mention one popular use case for each.

Ans: 1. Python

Use Case: It is widely used for Artificial Intelligence (AI) and data analysis because it is very easy to read and has powerful libraries.

2. JavaScript

Use Case: It is mainly used in web development to make websites interactive and build dynamic user interfaces that run in the browser.

3. Java

Use Case: It is very popular for building Android mobile applications and large-scale software systems for companies.

4. C++

Use Case: Because it is very fast and powerful, it is heavily used for developing high-end video games and system software.

5. SQL(Structured Query Language)

Use Case: It is specifically used for managing, storing, and retrieving data inside relational databases.

Q5)What is the difference between a developer and a tester? Answer in 2 lines.

Ans: A Developer creates and maintains software by writing code, adding features, and fixing coding-related problems.

A Tester checks the software to find bugs, verify that features work correctly, and make sure the software meets the required expectations.

Session – 2

Q1) What are the three main layers of software architecture? Write one line about each.

Ans:

1. Presentation Layer – This is the user interface layer where users interact with the application.

2. Business Logic Layer – This layer contains the main rules and logic that process user requests and perform operations.

3. Data Access Layer – This layer communicates with the database to store, retrieve, update, and delete data.

Q2) Define client and server in your own words with one real-life example.

Ans:

Client – A client is a device or application that sends a request to a server to get some service or information.

Server – A server is a computer or system that receives requests from clients, processes them, and sends back the required information.

Real-life Example – When we open YouTube on a mobile phone, the mobile app acts as the client and YouTube's servers process the request and send the videos and other data back to the phone.

Q3) Write 2 differences each between desktop apps, web apps, and mobile apps.

Ans:

1. Desktop Applications:

- * They are installed directly on a computer or laptop.
- * They can often work without an internet connection.

2. Web Applications:

- * They are accessed through a web browser such as Chrome, Edge, or Firefox.
- * They usually require an internet connection and do not need to be installed on the computer.

3. Mobile Applications:

- * They are designed specifically for mobile devices such as smartphones and tablets.
- * They can use mobile features such as GPS, camera, notifications, and sensors.

Q4) Name any 3 databases and mention whether each is SQL or NoSQL.

Ans:

1. MySQL – SQL Database
2. PostgreSQL – SQL Database
3. MongoDB – NoSQL Database

Session – 3

Q1) Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

Ans:

1. HTTP – HyperText Transfer Protocol

Use: It is used to transfer web pages and other data between a web browser and a web server.

2. HTTPS – HyperText Transfer Protocol Secure

Use: It is the secure version of HTTP that encrypts data transferred between the browser and the web server.

3. FTP – File Transfer Protocol

Use: It is used to transfer files between a client computer and a server over a network.

4. SMTP – Simple Mail Transfer Protocol

Use: It is used to send emails from an email client to a mail server and between mail servers.

Q2) What is the difference between IPv4 and IPv6? Answer in 2 lines.

Ans:

IPv4 uses 32-bit addresses, such as 192.168.1.1, and provides about 4.3 billion possible addresses.

IPv6 uses 128-bit addresses, such as 2001:db8::1, and provides a much larger number of IP addresses.

Q3) Run nslookup google.com on your system and write down the IP address you get.

Ans:

Command: nslookup google.com

IPv4 address: 172.217.23.238

IPv6 address: 2a00:1450:400e:809::200e

Q4) What do these HTTP status codes mean: 200, 301, 404, 500, 403?

Ans:

1. 200 – OK: The request was successful and the server returned the requested information.
2. 301 – Moved Permanently: The requested resource has been permanently moved to a new URL.
3. 404 – Not Found: The requested page or resource could not be found on the server.
4. 500 – Internal Server Error: The server encountered an unexpected problem while processing the request.
5. 403 – Forbidden: The server understood the request but refuses to allow access to the requested resource.

Q5) Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

Ans:

DNS – Domain Name System – is a system that converts human-readable domain names into IP addresses.

1. Root (.) – The root is the top level of the DNS hierarchy and directs requests to the appropriate TLD servers.

2. TLD (.com) – TLD means Top-Level Domain. In www.google.com, “.com” is the TLD.

3. Domain (google) – “google” is the main domain name registered under the .com TLD.

4. Subdomain (www) – “www” is a subdomain or host name used to identify a particular service or server under the google.com domain.

Therefore, in www.google.com:

www → Subdomain/Host

google → Domain

.com → TLD

. → Root

Session – 4

Q1) List all 6 phases of SDLC in order with one line about each.

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Ans:

SDLC – Software Development Life Cycle

1. Planning – The project requirements, goals, resources, and schedule are identified.
2. Requirements Analysis – The team collects and analyzes what the users and business need from the software.
3. Design – The software structure, user interface, database, and overall system design are planned.
4. Development – Developers write the code and build the software according to the design.
5. Testing – The software is tested to find and fix bugs and make sure it works as expected.
6. Deployment and Maintenance – The software is released to users and regularly maintained, updated, and improved.

Q2) Write 3 differences between Agile and Waterfall models.

Ans:

1. Agile is an iterative model where development is done in small cycles, while Waterfall follows a fixed step-by-step process.
2. Agile allows changes in requirements during development, while Waterfall usually requires requirements to be finalized before development starts.
3. Agile involves frequent testing and customer feedback, while Waterfall generally performs testing after the development phase is completed.

Q3) Name the 3 main Scrum roles and write one line about the responsibility of each.

Ans:

1. Product Owner – Manages the product requirements and prioritizes the work in the Product Backlog.
2. Scrum Master – Helps the team follow Scrum practices and removes obstacles that affect the team's work.
3. Developers – Design, build, test, and deliver the working product increment during each Sprint.

Q4) What is a Sprint in Scrum? Mention its typical duration.

Ans:

A Sprint is a fixed period in Scrum during which the team works on selected tasks and creates a usable product increment.

Typical Duration: A Sprint usually lasts 1 to 4 weeks, with 2 weeks being very common.

Q5) What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

Ans:

1. To Do – Tasks that are ready to be started but have not been started yet.
2. In Progress – Tasks that the team is currently working on.
3. Done – Tasks that have been completed and meet the required conditions.
4. Backlog – A list of tasks, features, bugs, and requirements that may be worked on in the future.

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Session – 5

Q1) Write the Git command for each: initialize a repo, stage all files, commit with a message, push to remote, pull latest changes.

Ans:

1. Initialize a repository:

```
git init
```

2. Stage all files:

```
git add .
```

3. Commit with a message:

```
git commit -m "Initial commit"
```

4. Push to remote:

```
git push origin main
```

5. Pull latest changes:

```
git pull origin main
```

Q2) What is the difference between git fetch and git pull? Answer in 2 lines.

Ans:

git fetch downloads the latest changes from the remote repository but does not merge them into the current branch.

git pull downloads the latest changes and automatically merges them into the current branch.

Q3) Create a new GitHub repository, push any small text file from your local system, and share the repo link.

Ans:

Steps:

1. Create a new repository on GitHub.
2. Create a small text file, for example, notes.txt, in the local project folder.
3. Run these commands:

```
git init
```

```
git add .
```

```
git commit -m "Add text file"
```

```
git branch -M main
```

```
git remote add origin <your-github-repository-url>
```

```
git push -u origin main
```

Repository Link: Paste your actual GitHub repository link here after creating and pushing the repository.

Q4) What is the difference between a Fork and a Pull Request? Answer in 2 lines.

Ans:

A Fork creates a personal copy of another person's GitHub repository in your own GitHub account.

A Pull Request is a request to merge your changes from a branch or fork into the original repository.

Q5) Create a new branch named feature-test, make one commit on it, and merge it back to the main branch. Write the commands you used.

Ans:

1. Create and switch to the new branch:

```
git checkout -b feature-test
```

2. Add or modify a file and stage the changes:

```
git add .
```

3. Commit the changes:

```
git commit -m "Add feature test"
```

4. Switch back to the main branch:

```
git checkout main
```

5. Merge the feature-test branch:

```
git merge feature-test
```

6. Push the updated main branch:

```
git push origin main
```